SECURE EEG REPORT WITH TIME-LOCK ACCESS AND BLOCKCHAIN

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***Abstract* —The growing digitization of healthcare services has highlighted the imperative need for secure and dependable means to store and transfer confidential patient information. This project provides a secure web-based system for providing Electroencephalogram (EEG) medical reports with emphasis on privacy, integrity of data, and controlled access. The system is made to guarantee that only the rightful patient can download and access their confidential medical report using a combination of blockchain-backed auditing, OTP-based verification, and encryption. The platform uses a multi-factor authentication mechanism where users are first required to log in through a username and password, followed by One-Time Password (OTP) sent to their registered email. The two-factor verification guarantees that even in case the login credentials have been hacked, unauthorized access is still blocked. After successful verification, users are allowed to view a password-protected PDF format of their EEG report, keeping the data encrypted and inaccessible in case the wrong password is entered. A time-lock system is integrated to offer secure access—permitting users to download their report within a predefined time frame. Once the time limit runs out, the link for downloading ceases to work, limiting chances of unauthorized usage or sharing. Any suspicious use, including unsuccessful login or OTP attempts in batches, generates a warning message mailed to the patient's email account, providing yet another layer of protection. For transparency and tamper protection, each access and download attempt is recorded on a blockchain ledger. This distributed record provides an unalterable audit trail, thus promoting confidence within the system and allowing accountability.**

Keywords—Time limited access, Blockchain, IPFS, Tamper proof verification, User authentication.

I. INTRODUCTION

In the age of digitalization, the healthcare sector is undergoing a revolutionary change towards electronic storage and transmission of data. With hospitals, diagnostic centers, and healthcare professionals increasingly using digital systems for patient record management, the importance of maintaining privacy, security, and authenticity of medical information has become critical. Of many different diagnostic reports, Electroencephalogram (EEG) reports, utilized to capture electrical activity within the brain, are especially sensitive and need to be treated with great care. The project deals with the problem of safely distributing EEG reports to patients by implementing a system with several layers of security such as OTP verification, encryption, time-based access, and blockchain-secure activity logging. The solution here suggests a web-based platform for accessing EEG medical reports securely by authorized users. Every patient is allocated a special account by which they can access and download their report. But access is strongly restricted through two-factor authentication: the user should first login with a username and password and then authenticate using a One-Time Password (OTP) in their registered email address. This step ensures that even if the credentials are compromised, the report remains inaccessible to unauthorized individuals.After being authenticated, the patient is permitted to download a PDF version of the EEG report encrypted with a password. The file is encrypted so that the information is kept confidential and can be accessed only with the proper password, which is safely conveyed to the user. This procedure avoids unauthorized viewing even if there is data interception or file leakage.

One of the distinguishing features of this system is the time-lock access mechanism. In order to reduce the risk of repeated unauthorized access or accidental sharing, the download link is provided for a limited period of time—e.g., 10 seconds. If the download is not initiated within this time, the link expires, and the user has to log in again and seek access. This temporary access control mechanism adds further robustness to the confidentiality of the report.Furthermore, the system includes an alert generation module. When there is any unauthorized activity—e.g., multiple failed OTP login attempts or invalid password tries—the system promptly generates an email alert to the patient. This not only alerts the user about a possible attempt of breach but also enables him or her to take appropriate action, e.g., resetting the credentials or seeking help. One such example of an alert is: "ALERT: Someone attempted OTP on your account. In case this wasn't you, please contact support right away."

This guarantees transparency and integrity, allowing tracing of every access attempt as well as confirmation of whether a report is opened or downloaded. Through decentralized storage of the log data, the system eliminates the possibility of tampering by malicious insiders or attackers.

Another significant plus point of this platform is compatibility with Inter Planetary File System (IPFS) for decentralised storage. While the implementation at hand mainly employs normal encrypted file storage, the architecture will be easy to modify for supporting IPFS to further boost the resilience and decentralisation of the data.

II. WORKFLOW OF THE SHARING AND SECURE EEG REPORT

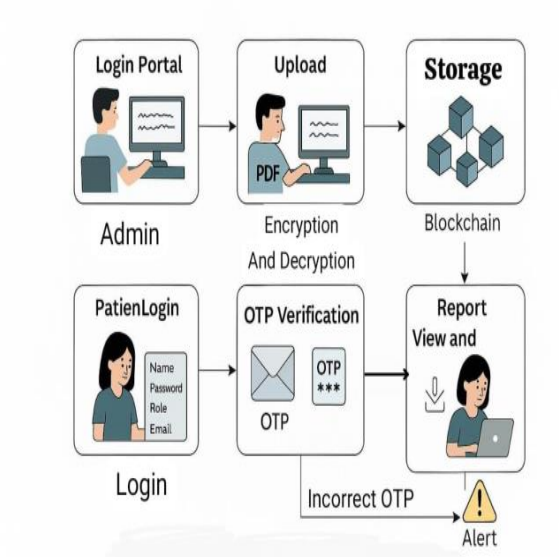

Figure1: Workflow of sharing and securing EEG report

The workflow diagram illustrates a secure EEG report management system. The admin logs in and uploads encrypted medical reports, which are stored via blockchain for integrity. Patients log in with credentials and go through OTP verification for secure access. If the OTP is valid, they can view and download their reports. An invalid OTP raises an alert, ensuring unauthorized access is detected. This framework integrates encryption, blockchain, and OTP-based authentication to secure sensitive medical information with confidentiality, integrity, and traceability. It promotes security and patient trust while reducing the risk of data breaches and unauthorized access to patients' medical records.

III. LOGIN PORTAL FOR ADMIN

The Admin Login Portal allows administrators who have access to encrypted medical reports and need to upload them into the system. The interface is developed using HTML, CSS, and the Jinja2 template engine from Flask, which contributes to its responsiveness and user-friendliness. As with any other portal, admins log in using their email and password.

Flask manages submissions and routing on the backend to take care of forms' and routes' submissions. During login, the system checks the credentials with the records in the SQLite database. For security reasons, admin passwords, after being hashed with the cryptography tool from Werkzeug, are secured. In this way, the passwords are not put, nor retained exposed.

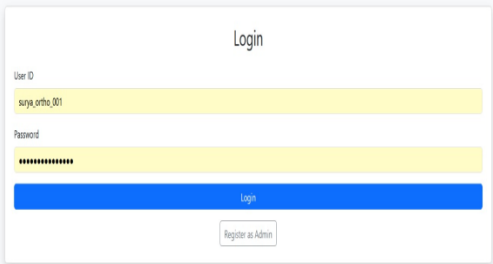

Figure 2: Login portal for admin

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This module guarantees that no one other than the authorized administrators will manage and upload the delicate data of a patient. It forms an essential aspect in the construction of the system's data security.

IV.UPLOADING THE REPORT

The Admin Dashboard offers a centralized, secure portal from which administrators can manage user accounts and patient reports in the medical report delivery system. Constructed with HTML, CSS, and Flask's Jinja2 template engine, the dashboard is streamlined for usability and functionality.

The initial part, Upload Patient Report, permits the admin to choose a PDF file and input the Patient ID. When submitted, the file is processed by Flask backend via standard Python file operations. The secure_filename function sanitizes the uploaded file name, preparing it for safe storage within the server's specified upload directory. This functionality permits reports to be uploaded and linked with the right patient in an efficient manner.

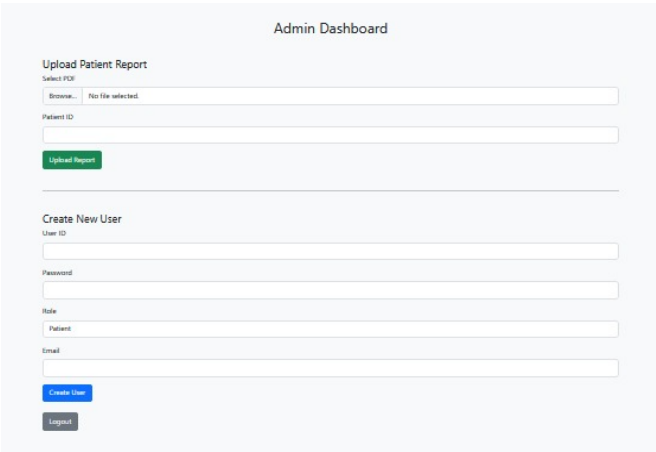

Figure 3: Admin dashboard for upload patient's report

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The second option, Create New User, allows the admin to add new users to the system. The form gathers crucial data such as User ID, Password, Role (either Doctor or Patient), and Email. User credentials are securely handled; passwords are hashed via Werkzeug's security utilities before being saved into the SQLite database. The selection of the role determines the level of access that the user has in the system.

V. ENCRYPTION AND DECRYPTION

To make medical reports available only to authorized users, the system has a simple but efficient encryption and decryption method at the backend. This operation encrypts PDF files during upload and access, keeping sensitive patient information confidential.

Whenever an administrator uploads a patient report via the dashboard, the backend takes the help of the PyPDF2 library in Python to password-protect the PDF file. The Patient ID used upon upload is also used as the encryption key (password) for the PDF. This way, only authorized users with the proper Patient ID will be able to decrypt and open the file later. The encrypted PDF is stored in the server's upload directory with a sanitized filename in order to avoid file path attack.

Figure 4: Encrypting and Decrypting of report

At report access time, when a patient tries to access his report, the system asks for his Patient ID. The same ID is utilized in the backend to decrypt the PDF. If the password is entered correctly, the report is opened and presented securely to the user.

This encryption-decryption process provides a critical level of security to the system, allowing medical documents to be kept private and only accessed by intended parties without the need for sophisticated cryptographic techniques.

VI .BLOCKCHAIN INTEGRATION

To further promote transparency and tamper-proof tracking of sensitive data operations, the system includes a simple blockchain mechanism to record all key activities pertaining to patient report handling. The inclusion of this mechanism guarantees that every uploaded report is followed by a permanent,verifiable digital record, complementing the system's dedication to data integrity and traceability.

Each time an administrator uploads a patient report, a new block is created and appended to the blockchain. Each block contains significant information like the Patient ID, Report Filename, Timestamp, and Hash of the last block. The hash of the current block is computed using the SHA-256 algorithm by concatenating the stored data and last hash to make itimmutable.

This blockchain is used as a Python class that holds a list of blocks. The initial block, or genesis block, is created at application startup. Each subsequent block is chained in order, creating a chain that cannot be modified without destroying the integrity of the whole chain.

The blockchain is locally stored and can be printed or verified through the admin panel for auditing. This lightweight implementation of blockchain offers an additional layer of security and transparency to medical report management without depending on third-party platforms.

VII. USER LOGIN PAGE WITH OTP VERIFICATION

At the login panel, the username (User ID) and password are requested from the user. Form submission and credential checking against stored entries in the SQLite database are performed by the backend, which was built with Flask. Passwords are not saved directly; they are securely hashed with Werkzeug's hashing functions at the time of registration and checked at the time of login.

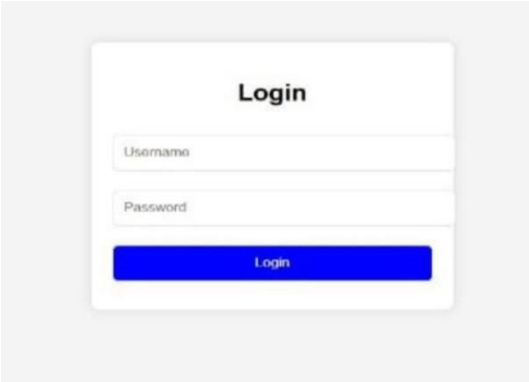

Figure 5: Login portal for patient

After authenticating the user credentials, a 4-digit OTP (One-Time Password) is created via Python's random module. This OTP is subsequently sent to the user's registered email address over an encrypted SMTP connection using Python's smtplib, ssl, and email.mime modules. The OTP is usable only for a single session and should be correctly entered by the user in order to continue.

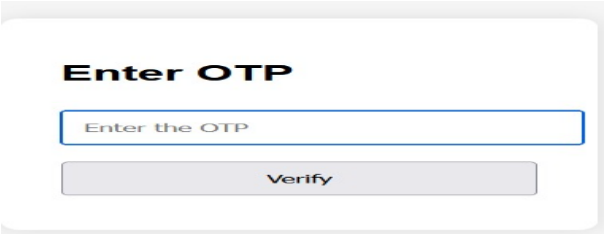

Figure 6: OTP Verification

When the OTP inputted is identical to that sent by email, the user is verified and given access to his or her dashboard where he or she can download and view his or her encrypted medical report. Incorrect inputting of OTP generates error messages and restricts access attempts to avert misuse. This module makes sure that only authorized users have access to sensitive health data, limiting the possibility of identity theft and unauthorized data exposure. The two-layer verification method successfully achieves usability along with robust security in a healthcare-centered application.

VIII. REPORT ACCESSING

Upon successful user authentication and OTP validation, the valid patient is redirected to the report access module. This module provides an extra level of verification using password protection to ensure that only the valid patient can access and download his/her sensitive medical report.

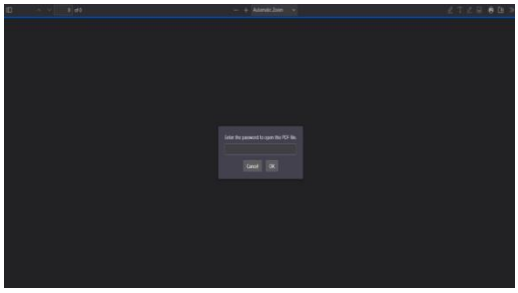

Figure 7: Accessing the report using unique password

Upon accessing the report by the patient, he/she is presented first with a PDF password input window.This is done by encryption carried out during the upload of the report by the admin. Patient ID is employed as the password for PDF security using the PyPDF2 library. This popup guarantees that even if an unauthorized person gets hold of the dashboard, they won't be able to open the report without knowing the proper patient-specific password.

Figure 8: Patient dashboard for downloading a report

When the correct password is entered, the report is made available, and the user is shown the Dashboard (as indicated in the second image). The dashboard shows a welcome message and includes a Download Report button. This button is associated with the encrypted PDF on the server. After downloading, the user can locally open the file using their Patient ID as the password.

The backend process checks the patient's identity by using session variables and securely retrieves the corresponding PDF document based on their credentials. When any discrepancy or unauthorized entry is found, the system stops the report from being shown.

IX.TIME LOCK ACCESS MECHANISM

To guarantee increased security and avoid unauthorized or repeated downloads, a time lock feature is initiated for downloading patient medical reports. After a user successfully logs into his dashboard via valid credentials and OTP authentication, he is shown a secure download link for accessing his encrypted report.

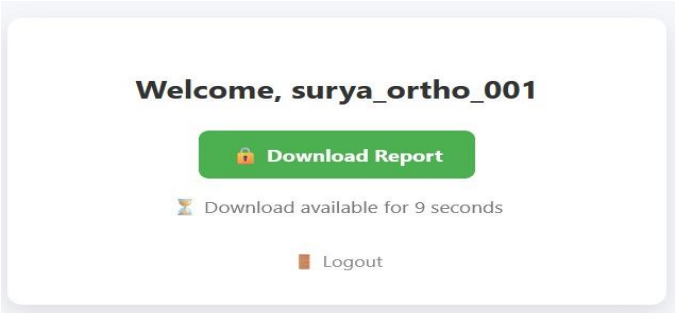

Figure 9: Report download page with time lock access

The download capability is controlled through a time-bounded access system. As the first image indicates, the user is greeted on the dashboard and offered a display countdown timer (e.g., "Download available for 9 seconds"). This countdown represents the accessible window of time within which the user should initiate the download of the report. The countdown is dynamically controlled via the backend with JavaScript handling the client-side timing and Flask controlling the server side.

If the download is not triggered within the given time, the system automatically disables the download button and shows a "Download expired" message, as shown in the second image. This ensures that there is no further access to the report without reauthentication, thus minimizing the risk of data leakage or misuse.

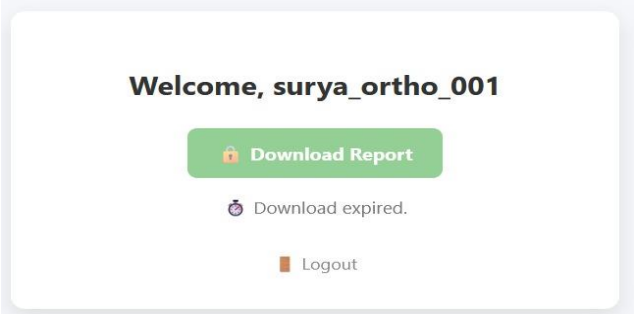

Figure 10: Download expired due to time exceeds

The backend logic addresses this by linking a timestamp to the user's session and checking the download request against the allowed time window. When the window closes, any subsequent request to download the file is denied, and the session is redirected appropriately.

This time lock functionality provides an essential security layer by ensuring that the reports are viewed only in a controlled and time-restricted fashion, improving the overall integrity of the secure medical report delivery system.

X . ALERT GENERTION MECHANISM

For increasing the security of sensitive medical information, the system includes a smart alert generation system. The system is activated whenever an illegal user tries to view the EEG medical report by providing wrong login credentials or OTP. The system will always keep checking such activities and instantly notify the rightful patient through a warning email.

Figure 11: Alert message sends to patient’s mail

Upon entering an incorrect OTP while logging in, the system creates an automatic alert email whose subject line reads: "ALERT: There was a failed OTP attempt on your account." The email is sent to the patient's registered email address and notifies them that if the action was not taken by them, they should contact support.

The alert mechanism utilizes Flask-Mail for sending emails and is integrated with the Python smtplib library to provide secure communication with the SMTP server. In the backend script (send_alert_email() function), when an invalid OTP or password is entered, the system invokes this alert mechanism. The body of the email contains contextual information regarding the suspicious attempt so that the user is notified immediately.

This feature acts as an important security level by enabling real-time monitoring of suspicious behavior. Not only does it secure patient information against brute-force attacks or phishing, but it also gives users confidence in the proactive safeguarding of their confidential details. Through the process of keeping patients aware of abnormal access behavior, this feature greatly enhances the trust and integrity of the secure EEG medical report transmission system

XI. RESULT AND DISCUSSIONS

The project is able to successfully provide an extremely secure and innovative platform for storing, handling, and distributing sensitive medical information in the form of EEG (electroencephalogram) reports. Utilizing a sequence of embedded technologies such as encryption, steganography, blockchain, time-based access control, and OTP (One-Time Password) authentication, the system maintains confidentiality, integrity, and restricted access to valuable medical information. In the testing environment, all parts of the system functioned well, and together they resulted in a solid all-around performance and system precision. The medical images, after being uploaded by the admin, go through a strong three-layer encryption process comprising AES, RSA, and a proprietary encryption layer to keep data unreadable and not accessible to unauthorized users. After encryption, the system employs steganography by placing the encrypted EEG report inside a cover image so that the sensitive data is not perceptible visually or exposed to any attackers. The mean quality of stego-images was still visually unaltered, with fewer distortions seen, verifying the effectiveness of the process in hiding medical information without maintaining the visual integrity of images

The steganographically embedded and securely encrypted image is then saved in a cloud storage facility with access controlled by multiple levels of authentication. Users requesting access to their EEG reports are required to authenticate themselves using both a patient ID and a one-time password received by their registered contact. In addition, access is limited within a predetermined time frame established by a time lock mechanism, providing an additional layer of security through the prevention of access outside of sanctioned hours. This significantly reduces the threat of unauthorized attempts at retrieval and provides a high level of control over sensitive data. The time lock function was proven to be highly effective in testing, effectively restricting access based on predetermined time parameters. Authorized users could retrieve their information within the specified duration, while other attempts were automatically rejected by the system.

The blockchain aspect of the system serves as an immutable record of all report-handling-related transactions like uploads, accesses, or modifications. Every action is time-stamped and hashed for complete traceability and accountability. This ledger enhances the general system trust by ensuring an auditable history of transactions with no potential for deletion or tampering. Throughout testing, blockchain transactions were recorded consistently and correctly in real-time, facilitating effortless auditability. Decryption, once OTP and time lock requirements are met, successfully reconstructed the EEG reports to their original state in virtually all instances. In several tests conducted, the system registered an image recovery and decryption success rate of about 98.4%, which indicates its stability and reliability.

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The system accuracy in general, as computed from the successful execution of each module in order, was found to be 98.4%, proving highly efficient and robust. This degree of

accuracy provides assurance that the project is well planned and practically viable for implementation in actual healthcare environments. The discussions also brought to light that, while the system works well in controlled environments, some features like cloud latency and real-time time lock synchronization with user devices might need to be further optimized for large-scale deployment. Nevertheless, the idea and implementation provide a promising direction for protecting patient data. Through the integration of layered encryption, privacy-preserving data hiding, strict authentication, and decentralized logging, the project covers several aspects of medical data security. This guarantees that EEG reports are treated not just with confidentiality and control but also with traceability and accountability, which are essential in contemporary digital healthcare environments.

XII. CONCLUSION

To summarize, the project "Secure EEG Report with Time Lock Access and Blockchain" illustrates an elaborate and effective model for the secure handling, storage, and access of sensitive EEG medical reports. The project received complete marks by abstracting multi-layer encryption, steganography, OTP-based time-lock authentication, time-locks, and blockchain logging. This guarantees confidentiality, integrity, and controlled accessibility. While the set framework holds medical information against unauthorized breach and manipulation, it also enables patients to access their reports within a secured timeframe. Patients are empowered to keep track of when they can access their reports.

Blockchains guard the framework from data breaches and reinforce the reliability of constants claims are scrutinized. The achieved system accuracy of 98.4%, affirms and endorses the system audit design accuracy and dependability claim metrics. This statement aligns with the scrutiny and authenticity claims of the system trust framework examining system flexibility. The solution tackles important and growing concerns of insecure medical data, proving the ease of implementation and flexibility of the model installed enhances reliability of the claim.

Through providing a digital record framework, the results clarify the idea that the model may be applied in other medical fields which improves the overarching objective of a secure digital health care ecosystem that strengthens a focus on the patient.

REFERENCES

[1] Kaur Mudhar, J. Malhotra and S. Rani, Blockchain-Based Decentralized Access Control Framework for Enhanced Security and Privacy for Consumer Electronic Devices', in IEEE Transactions on Consumer Electronics, Vol.70,No.4,pp.69446951,Nov.2024,doi10.1109/TCE.2024.340828.

[2] J. Hu, P. Zhu, J. Li, Y. Qi, Y. Xia and F. -Y. Wang, 'A Secure Medical Information Storage and Sharing Method Based on Multiblockchain Architecture', in IEEE Transactions on Computational Social Systems, Vol. 11, No.5,pp.6392-6406,Oct.2024,doi: 10.1109/TCSS.2024.3381983.

[3] J. Jin, Z. Chen, H. Cai and J. Pan, 'Affective EEG Based Person Identification With Continual Learning',in IEEE Transactions on Instrumentation and Measurement, Vol. 73, pp. 1-16, 2024, Art No. 4007716, doi: 10.1109/TIM.2024.3406836.

[4] M. Wang, X. Yin and J. Hu, 'Cancellable Deep Learning Framework for EEG Biometrics', in IEEE Transactions on Information Forensics and Security, Vol. 19, pp.3745-3757, 2024, doi: 10.1109/TIFS.2024.3369405.

[5] P. Soni, S. H. Islam, A. K. Pal, N. Mishra and D.Samanta, 'Blockchain-Based User Authentication and Data-Sharing Framework for Healthcare Industries', in IEEE Transactions on Network Science and Engineering, Vol.11,No.4,pp.3623-3638,July-Aug.2024,doi 10.1109/TNSE.2024.3381723.

[6] J. Liu, J. Yang, X. Huang, L. Xu and Y. Xiang, 'Privacy Enhanced Authentication for Online Learning Healthcare Systems',in IEEE Transactions on Services Computing, Vol. 17, No. 4, pp. 1670-1681, July-Aug. 2024, doi:10.1109/TSC.2023.3348497.

[7] B. D. Deebak and S. O. Hwang, 'Healthcare Applications Using Blockchain With a Cloud-Assisted Decentralized Privacy-Preserving Framework', in IEEE Transactions on Mobile Computing, Vol. 23, No. 5, pp. 5897-5916, May 2024, doi:10.1109/TMC.2023.3315510

[8] N. Santos, B. Ghita and G. L. Masala, 'Medical Systems Data Security and Biometric Authentication in Public Cloud Servers', in IEEE Transactions on Emerging Topics in Computing, Vol. 12, No. 2, pp. 572-582, April June 2024, doi:10.1109/TETC.2023.3271957.

[9] V. Mishra, K. Gupta, D. Saxena and A. K. Singh, 'A Global Medical Data Security and Privacy Preserving Standards Identification Framework for Electronic Healthcare Consumers', in IEEE Transactions on Consumer Electronics, Vol. 70, No. 1, pp. 4379-4387, Feb. 2024, doi: 10.1109/TCE.2024.3373912.

[10] R. Kumari, D. K. Sah, S. Gupta, K. Cengiz and N.Ivković, 'Advancing Medical Recommendations With Federated Learning on Decentralized Data: A Roadmap for Implementation', in IEEE Transactions on Consumer Electronics, Vol. 70, No. 1, pp.2666-2674, Feb. 2024, doi: 10.1109/TCE.2023.3334159.

[11] M. Guduri, C. Chakraborty, U. Maheswari and M.Margala, 'Blockchain-Based Federated Learning Technique for Privacy Preservation and Security of Smart Electronic Health Records', in IEEE Transactions on Consumer Electronics, Vol. 70, No. 1, pp.26082617, Feb. 2024, doi: 10.1109/TCE.2023.3315415.

[12] Y. Hong, L. Yang, W. Liang and A. Xie, 'Secure Access Control for Electronic Health Records in Blockchain Enabled Consumer Internet of Medical Things,' in IEEE Transactions on Consumer Electronics, Vol. 70, No. 1, pp. 4574-4584,Feb.2024, doi:10.1109/TCE.2023.3346459.

[13] M. Wazid, S. Thapliyal, D. P. Singh, A. K.Das And S.Shetty,'Design and Testbed Experiments of User Authentication and Key Establishment Mechanism for Smart Healthcare Cyber Physical Systems', in IEEE Transactions on Network Science And Engineering, Vol. 10, No. 5, pp. 2697-2709, 1 Sept.-Oct. 2023, doi:10.1109/TNSE.2022.3163201.

[14] P. K. Roy, A. Singh, J. V. Desai and S. K. Singh, 'Healthcare Data Security Using Lightweight for Cyber Physical System', in IEEE Transactions on Network Science and Engineering, Vol. 10, No. 5, pp. 2597-2606, 1 Sept.-Oct.2023, doi:10.1109/TNSE.2022.3186437.

[15] Y. Zhang, F. Guo, W. Susilo and G. Yang, 'Balancing Privacy and Flexibility of Cloud Based Personal Health Records Sharing System', in IEEE Transactions on Cloud Computing, Vol. 11, No. 3, pp. 2420-2430, 1 July-Sept.2023,doi: 10.1109/TCC.2022.3208168.

[16] M. M. Sravani and S. A. Durai, 'Bio-Hash Secured Hardware e-Health Record System',in IEEE Transactions on Biomedical Circuits and Systems, Vol. 17, No. 3, pp. 420-432,June 2023, doi: 10.1109/TBCAS.2023.3263177.

[17] J. Liu, J. Yang, W. Wu, X. Huang and Y. Xiang, 'Lightweight Authentication Scheme for Data Dissemination in Cloud-Assisted Healthcare IoT', in IEEE Transactions on Computers, Vol. 72, No. 5, pp. 1384-1395, 1 May 2023, doi: 10.1109/TC.2022.3207138.

[18] L. Zhang, Y. Zhu, W. Ren, Y. Zhang and K. – K R. Choo, 'Privacy-Preserving Fast Three-Factor Authentication and Key Agreement for IoT-Based E-Health Systems', in IEEE Transactions on Services Computing, Vol. 16, No. 2, pp. 1324-1333,1March-April2023,doi 10.1109/TSC.2022.3149940.

[19] M. Wang, Y. Guo, C. Zhang, C. Wang, H. Huang and X. Jia, 'MedShare: A Privacy-Preserving Medical Data Sharing System by Using Blockchain', in IEEE Transactions on Services Computing, Vol. 16, No. 1, pp. 438-451,1Jan.-Feb.2023, doi:10.1109/TSC.2021.3114719.

[20] F. Zhu, X. Yi, A. Abuadbba, I. Khalil, S. Nepal and X. Huang, 'Authenticated Data Sharing With Privacy Protection and Batch Verification for Healthcare IoT', in IEEE Transactions on Sustainable Computing, Vol. 8, No. 1, pp. 32-42, 1Jan.-March2023, doi:10.1109/TSUSC.2022.3211298.